Sergi Martínez

Robotics PhD researcher | System identification, dynamics and control

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PhD viva: 16 November 2026 | Available to start immediately

Research profile

Robotics PhD researcher specialising in physically consistent system identification, contact dynamics and optimal control for manipulation and locomotion. Develops structured estimation algorithms and C++/Python research software, connecting mathematical methods to simulation and robot experiments. First-author research in *IEEE Transactions on Robotics* (conditionally accepted) and *IEEE Robotics and Automation Letters*, complemented by experience in cloth manipulation and robot integration.

Selected publications

1. **S. Martinez**, S. Tonneau and C. Mastalli. "System Identification under Constraints and Disturbance: A Bayesian Estimation Approach." *IEEE Transactions on Robotics*, **conditionally accepted**, 2026. [Video]
2. **S. Martinez**, R. J. Griffin and C. Mastalli. "Multi-Contact Inertial Parameters Estimation and Localization in Legged Robots." *IEEE Robotics and Automation Letters*, 10(5):4730–4737, 2025. [Video]
3. M. Parilli, **S. Martinez** and C. Mastalli. "Endpoint-Explicit Differential Dynamic Programming via Exact Resolution." *IEEE International Conference on Robotics and Automation (ICRA)*, 2025. [Video]
4. J. Rojas, A. Papatheodorou, **S. Martinez**, A. Patrizi, I. Havoutis and C. Mastalli. "ODYN: An All-Shifted Non-Interior-Point Method for Quadratic Programming in Robotics and AI." Submitted to *IEEE Transactions on Robotics*; under review, 2026.

Doctoral research

PhD researcher in Robotics, Heriot-Watt University 2022–present

Robot Motor Intelligence (RoMI) Lab, Edinburgh | Supervisor: Carlos Mastalli

- **Constrained system identification [1]:** Developed a Bayesian framework that jointly estimates robot states, inertial parameters and friction while enforcing contact and loop-closure constraints. Derived a structured Riccati recursion with backward-pass complexity linear in the trajectory horizon.
- **Parameter observability [1]:** Incorporated energy observations to separate inertial and friction effects. Reduced friction-parameter error by approximately 50% in the biped and quadruped simulation ablation, averaged over 100 random initialisations per model.
- **Hardware validation [1]:** Identified physically consistent models on a Unitree B1 with a Z1 arm, including the legs' four-bar mechanisms and joint friction. Improved end-effector tracking against the nominal URDF model in standing manipulation and validated the model in stair-climbing experiments.
- **Multi-contact estimation [2]:** Developed a multiple-shooting solver for payload identification and robot localisation, with physically consistent inertial parametrisation and analytical contact-dynamics derivatives. Evaluated against least-squares estimation and validated experimentally on a Go1 robot.

Technical skills

Programming and software: C++, Python, ROS, Crocoddyl, Pinocchio, URDF; code generation, profiling and numerical testing.

Methods: Rigid-body and contact dynamics, state and parameter estimation, nonlinear MPC, differential dynamic programming, trajectory optimisation, differentiable simulation and QP layers.

Experiments: Robot and sensor integration, force/torque sensing, signal processing, mechanism prototyping and hardware validation.

Research software

Crocoddyl | Joint state and parameter estimation
Develop parameter-aware dynamics, costs, constraints and observers in C++/Python, with analytical derivatives, numerical differentiation checks and solver tests. Public implementation available as draft pull request #1525.

ODYN | Differentiable numerical optimisation
Contribute differentiable QP layers, code generation, solver testing and performance-oriented C++ for robotics and learning. Co-author of the associated solver paper.

GeneSys | Differentiable contact simulation
Contribute to differentiable contact-dynamics simulation and its integration with optimisation and control.

Research and engineering experience

Research Support Engineer, IRI (CSIC–UPC) May 2019–Nov 2022
Perception and Manipulation Group, Barcelona

- Developed C++/ROS algorithms and integrated robot software, sensors and hardware for cloth manipulation in the ERC CLOTHILDE project. Maintained and extended research software to support manipulation experiments.

Research Intern, Delft University of Technology Oct 2018–May 2019
Precision and Microsystems Engineering, Delft

- Designed, built and tested a dynamically balanced planar translator. Used nonlinear compliant-mechanism models to guide the mechanical design and experiments to validate the balancing approach.

Research Intern, IRI (CSIC–UPC) Sep 2018–Sep 2019
Computational Robotics, Barcelona

- Instrumented a 6-DoF Gough–Stewart platform as a force/torque sensor and developed signal processing for variable-compliance behaviour.

Education

PhD in Robotics, Heriot-Watt University 2022–present
Viva scheduled: 16 November 2026 | Edinburgh, UK

MSc in Automatic Control and Robotics, UPC 2020–2022
Universitat Politècnica de Catalunya, Barcelona | GPA: 8.02/10

Developed nonlinear MPC for aggressive UAV manoeuvres, incorporating actuator dynamics into the model. Extended Crocoddyl's DDP solver and benchmarked against a conventional model in simulation.

Dual BTech in Mechatronics and Industrial Electronics 2014–2019
University of Vic – Central University of Catalonia, Spain

Degrees in Mechatronics Engineering and Industrial Electronics Engineering and Automation.

Recognition

EPS Postgraduate Research 2nd Year Prize, 2025; second-best paper award, ICCMA 2019.